

$$\approx 0.58 \text{ m/s}$$

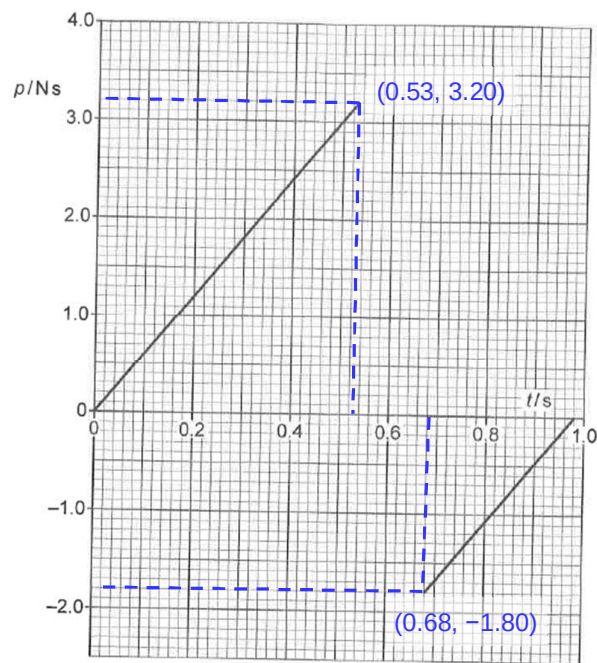
**2(a)**

From Fig. 2.1, momentum of ball as it makes contact with the surface,  
 $p = 3.2 \text{ N s}$

Using  $p^2 = 2mE_k$ ,

$$\begin{aligned} E_k &= \frac{p^2}{2m} \\ &= \frac{(3.2)^2}{2(0.62)} \\ &= 8.258 \\ &\approx 8.26 \text{ J} \end{aligned}$$

**2(b)**



**Fig. 2.1**

Show clear construct working on Fig. 2.1.

Taking upwards as positive,

$$\begin{aligned} \Delta p &= 1.80 - (-3.20) \\ &= 5.00 \text{ N s} \end{aligned}$$

| No.         | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>By Newton's 2nd law,</p> $\langle F \rangle = \frac{\Delta P}{\Delta t}$ $= \frac{5.00}{(0.68 - 0.53)}$ $= \frac{5.00}{0.15}$ $= 33.33$ $\approx 33.3 \text{ N}$ <div data-bbox="483 659 850 890" data-label="Diagram"> </div> <p>By Newton's second law,</p> $F_{\text{resultant}} = N - W$ $N = F_{\text{resultant}} + W$ $= 33.33 + (0.62)(9.81)$ $= 39.41$ $\approx 39.4 \text{ N}$ <p><i>Note: A common error is made by not taking the direction change into account.</i></p> |
| <b>2(c)</b> | <p>Kinetic energy of ball immediately after 1st rebound,</p> $E_1 = \frac{p^2}{2m}$ $= \frac{(1.8)^2}{2(0.62)}$ $= 2.613 \text{ J}$ <p>Fraction of kinetic energy of ball after each rebound,</p> $\frac{E_1}{E_0} = \frac{2.613}{8.258} = 0.3154$ $\frac{E_2}{E_1} = 0.3154$ $E_2 = 0.3154 E_1$ $= (0.3154)^2 E_0$                                                                                                                                                                    |

| No.  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>After <math>n</math>th rebound,</p> $E_n = (0.3154)^n E_0$ <p>For total energy to decrease to 5.0% of its initial energy,</p> $(0.3154)^n E_0 = 0.050 E_0$ $n = \frac{\log(0.050)}{\log(0.3154)}$ $= 2.60$ <p>Thus, number of rebounds to decrease to less than 5.0% of its initial energy = 3</p> <p><i>Note: A common error is not counting the first bounce, or rounded the number bounces down, hence incorrectly reaching two bounces as an answer.</i></p> |
| 3(a) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |